

RoCBF-Net: A Tunable Safety Architecture for High-Order CBFs Under Uncertainty—GP Correction, Tunable Margins, and the Deployment Envelope

Zhuang Shao^{1,*}, Lijun Lei², Peng Wang³, Liang Zheng², Jie Zhou²

¹China Resources Power Technology Research Institute Co., Ltd.,
Shenzhen 518000, China

²Rundian Energy Science and Technology Co., Ltd.,
Zhengzhou 450052, China

³North China University of Water Resources and Electric Power,
Zhengzhou 450046, China

*Corresponding author: shaozhuang@crpower.com.cn,
ORCID: 0000-0003-2496-0797

Abstract

Safety filters based on control barrier functions (CBFs) project control actions into the safe set, but their certificate relies on accurate dynamics—a condition routinely violated in industrial plants under coal quality drift, equipment aging, and unmodeled disturbances. This paper presents RoCBF-Net, a tunable safety filtering architecture that combines three components: high-order CBF (HOCBF) constraints with correct relative degrees, Gaussian process (GP) residual learning, and a scalar robustness factor ϵ_κ that scales the compositional uncertainty margin in the quadratic program constraint. The key finding is that GP mean correction—not the compositional $\epsilon(x)$ margin—is the dominant safety mechanism, reducing constraint violations by $> 500\times$ versus uncorrected HOCBF across eight conditions on a 5th-order supercritical boiler-turbine benchmark. Moreover, the optimal ϵ_κ depends systematically on the uncertainty structure: $\epsilon_\kappa = 0$ suffices for additive bounded perturbations, while $\epsilon_\kappa = 0.1$ is required for state-dependent coupling (recovering zero violation from 33.5% failure). Crucially, the theoretical worst-case $\epsilon_\kappa = 1.0$ is never the optimal choice under any tested condition, and a coupling-strength sweep reveals a *deployment envelope*—the safe ϵ_κ range collapses from $[0.1, 1.0]$ at weak coupling to the singleton $\{0\}$ at strong coupling ($\gamma \geq 1.5$), where

any positive margin harms safety. These results motivate treating ϵ_κ as a tunable calibration parameter selected per uncertainty type and strength, rather than a fixed theoretical constant. The safety-filter pipeline runs at ~ 25 ms per step (GPU, JIT-compiled), approximately ten times faster than the implemented SLSQP-based NMPC baseline.

Keywords: Control barrier functions; safety filtering; Gaussian processes; model mismatch; robustness margin calibration; deployment envelope; supercritical power plant

1 Introduction

1.1 Industrial Measurement and Control Context

Supercritical thermal power units—the backbone of modern electricity generation—operate at main steam pressures exceeding 22 MPa and temperatures above 600°C. Safe operation requires continuous measurement and constraint-preserving control of three critical process variables: main steam pressure, separator outlet enthalpy, and turbine power output. These constraints originate from multiple layers of the industrial control architecture: sensor measurement ranges, actuator saturation limits, equipment protection boundaries, and coordinated control requirements.

During daily operation, the nominal plant model diverges from reality. Coal quality fluctuates across shipments, heat exchanger surfaces foul progressively, and actuators degrade over months of service. These factors introduce persistent model mismatch between the design model used for controller synthesis and the true plant dynamics. Under such mismatch, a controller—whether classical PID, model predictive control (MPC), or a learned policy—may issue control actions that appear safe under the nominal model but violate physical constraints on the actual plant.

1.2 The Safety Layer Approach

Rather than modifying or replacing the upstream controller, we adopt a *safety layer* architecture: an independent measurement-and-control safety filter that sits between any upstream controller and the plant actuators. The filter receives the controller’s proposed action u_{prop} , evaluates it against the current plant measurements and a calibrated model of the dynamics, and projects it to the nearest action u^* that is certified safe. This architecture has two practical advantages for industrial deployment: (i) it does not require modification of existing control logic, and (ii) the safety guarantee is independent of the upstream controller type—LQR, MPC, PID, or a trained policy all receive the same protection.

The technical mechanism for this projection is the control barrier function (CBF) [1], which enforces forward invariance of a safe set through a quadratic program (QP) solved

at each measurement sample. For industrial process variables with high relative degree—steam pressure requires two Lie derivatives of the constraint function to reach the control input—high-order CBFs (HOCBFs) [2] extend this construction through a recursive ψ -chain. The fundamental limitation is that under model mismatch, the HOCBF constraint is computed from incorrect dynamics, and the QP may certify an unsafe action as safe.

1.3 Technical Gap: Uncertainty Propagation in High-Relative-Degree Constraints

Gaussian processes (GPs) learn model residuals from measurement data with calibrated uncertainty quantification [3, 4]. Several works combine GPs with CBFs for robust safety [5, 6], but apply the GP uncertainty only at the top CBF level ($m = 1$). A recent concurrent work [7] extends this to HOCBFs via SOCP chance-constraint reformulation, providing a convex safety filter—but the reformulation is global rather than recursive, aggregating uncertainty at the constraint level rather than through each ψ -level individually. For industrial constraints with $m \geq 2$, this is structurally significant: the model residual propagates through successive Lie derivatives of the constraint function, and the uncertainty amplifies through each level of the ψ -chain. Our compositional σ -chain captures this per-level amplification by propagating per-dimension GP posterior uncertainties recursively, yielding a per-constraint, state-dependent robustness margin $\epsilon(x)$ that more precisely reflects the uncertainty structure at each relative-degree level. Similarly, cautious MPC [8] propagates GP uncertainty through a prediction horizon but at substantially higher computational cost and without per-step forward-invariance certification.

1.4 Contributions

We introduce RoCBF-Net, a tunable safety filtering architecture with four contributions:

1. **Tunable safety architecture (HOCBF + GP + ϵ_κ).** RoCBF-Net integrates high-order CBF constraints with correct per-constraint relative degrees, Gaussian process residual learning from commissioning data, and a scalar robustness factor ϵ_κ that scales the compositional uncertainty margin in the QP constraint $Au \leq b - \epsilon_\kappa \epsilon(x)$. The three components operate independently: GP can be configured as scenario-specific (pre-trained), online (incrementally adapted), or omitted; ϵ_κ is adjusted per uncertainty type and strength. Theorem 1 establishes that under a fixed, scenario-calibrated GP, the filter provides a high-probability forward-invariance certificate.
2. **GP mean correction as the dominant safety mechanism.** Through systematic ablation on a 5th-order supercritical boiler-turbine benchmark under Φ -scaled

nonlinear dynamics with eight conditions and eight method variants, we show that adding GP residual learning to the HOCBF constraint reduces violations by $> 500\times$ versus uncorrected HOCBF, while the full compositional $\epsilon(x)$ margin ($\epsilon_\kappa = 1.0$) degrades performance on additive-uncertainty conditions (S2, S4) compared to GP-only ($\epsilon_\kappa = 0$). This empirically establishes GP correction, not the robustness margin, as the primary driver of safety.

3. The deployment envelope: ϵ_κ depends on uncertainty type and strength.

An ϵ_κ sensitivity sweep across additive, state-dependent, and nonlinear perturbation scenarios reveals that the optimal ϵ_κ is not universal: $\epsilon_\kappa = 0$ suffices for additive bounded uncertainty, while $\epsilon_\kappa = 0.1$ is required for state-dependent coupling. Critically, $\epsilon_\kappa = 1.0$ (the theoretical worst-case bound) produces *worse* safety than a smaller value under *all* tested conditions. A coupling-strength gradient experiment further reveals a *deployment envelope*: the safe ϵ_κ range shrinks from $[0.1, 1.0]$ at $\gamma = 0.5$ to $[0.1, 0.5]$ at $\gamma = 1.0$, and collapses to the singleton $\{0\}$ at $\gamma \geq 1.5$ —beyond which any positive margin constrains the QP without providing protection.

4. Practical deployment methodology.

We characterize three GP calibration strategies (scenario-specific, online adaptation, and generic mixed GP with out-of-distribution failure mode) and provide an ϵ_κ selection schedule keyed to the expected perturbation structure and coupling strength. The method is intended as a commissioning-calibrated safety filter: characterize the uncertainty regime during commissioning, then configure ϵ_κ accordingly.

2 Related Work

2.1 Constrained Control and Safety Filtering

MPC is the dominant constrained control methodology in process industries [9, 10], with offset-free variants handling steady-state model mismatch [11]. Cautious MPC [8] incorporates GP uncertainty into the prediction horizon, providing chance-constrained safety at increased computational cost.

CBFs [1, 12] enforce forward invariance through QP-based control projection. HOCBFs [2] extend this to high-relative-degree constraints. Robust CBF formulations [13, 14] add margins based on structural disturbance bounds rather than data-driven estimates. A common practice across these works is to apply the full theoretical robustness margin—the worst-case bound derived from the analysis—without examining whether a smaller margin would suffice in practice. Our empirical results challenge this default: $\epsilon_\kappa = 1.0$ degrades safety under all tested conditions compared to a smaller calibrated value, and the optimal margin depends systematically on both the perturbation type (additive vs.

state-dependent) and its coupling strength.

2.2 GP-Enhanced CBFs

GPs provide calibrated uncertainty quantification for data-driven modeling [3, 4]. Several works combine GPs with CBFs [5, 6, 15], applying GP uncertainty at the top CBF level ($m = 1$). Aali & Liu [7] extend GP-HOCBF integration to $m \geq 2$ by converting the chance constraint into a second-order cone program (SOCP), yielding a convex safety filter. Zhang et al. [16] propose an adaptive fast variational sparse GP with HOCBF safety filtering for real-time robot control. These concurrent works differ from ours in two respects: (i) they apply GP uncertainty via global constraint reformulation rather than the recursive σ -chain propagation through each ψ -chain level, and (ii) they target robotics and numerical simulations rather than industrial process control with DCS deployment constraints. For industrial constraints with $m \geq 2$, our compositional σ -chain propagation through the full ψ -chain is structurally distinct. Lederer et al. [17] derive uniform GP error bounds complementary to our approach.

2.3 Safety Margin Calibration

A gap in the existing literature is the systematic empirical characterization of when robustness margins help versus harm. Prior work on CBF robustness margins treats the margin as a certificate to be satisfied—scaled at the worst-case bound—rather than a deployment parameter to be calibrated. We provide the first experimental demonstration that the GP-CBF robustness factor ϵ_κ has a *deployment envelope*: a bounded regime of coupling strengths and perturbation types within which a positive margin is necessary and effective, bounded above by a regime where it becomes counterproductive. This reframes ϵ_κ from a theoretical constant into a commissioning-calibrated design parameter.

In the broader safety filtering landscape, shielding approaches [18] provide safety by correcting policy actions, and constrained RL methods [19] enforce expected constraints. Adaptive CBFs [20] estimate uncertain parameters online. Our decoupled design provides a model-based, per-step probabilistic certificate compatible with any upstream controller.

3 RoCBF-Net: Architecture and Theory

3.1 Problem Formulation

Consider a control-affine system with model mismatch:

$$\dot{x} = \underbrace{f_0(x) + g_0(x)u}_{\text{nominal}} + \underbrace{\Delta f(x)}_{\text{residual}}, \quad x \in \mathbb{R}^n, \quad u \in \mathcal{U} \subset \mathbb{R}^m \quad (1)$$

where Δf represents unmodeled dynamics, parameter drift, and external disturbances. We assume the input matrix is known ($\Delta g = 0$), as actuator characteristics are typically well-characterized in industrial settings. For scenarios where actuator degradation primarily affects process variables through reduced throughput (e.g., valve fouling reducing steam extraction efficiency) rather than through changes in actuator gain, the dominant effect is captured by Δf ; the benchmark models actuator degradation scenarios (S5) in this form.

For a constraint $h(x) \geq 0$ with relative degree m , the HOCBF ψ -chain is:

$$\psi_0 = h(x), \quad \psi_i = \dot{\psi}_{i-1} + \alpha_i(\psi_{i-1}), \quad i = 1, \dots, m \quad (2)$$

With linear $\alpha_i(r) = k_i r$, the constraint $A_0(x)u \leq b_0(x)$ enforces forward invariance when the model is exact.

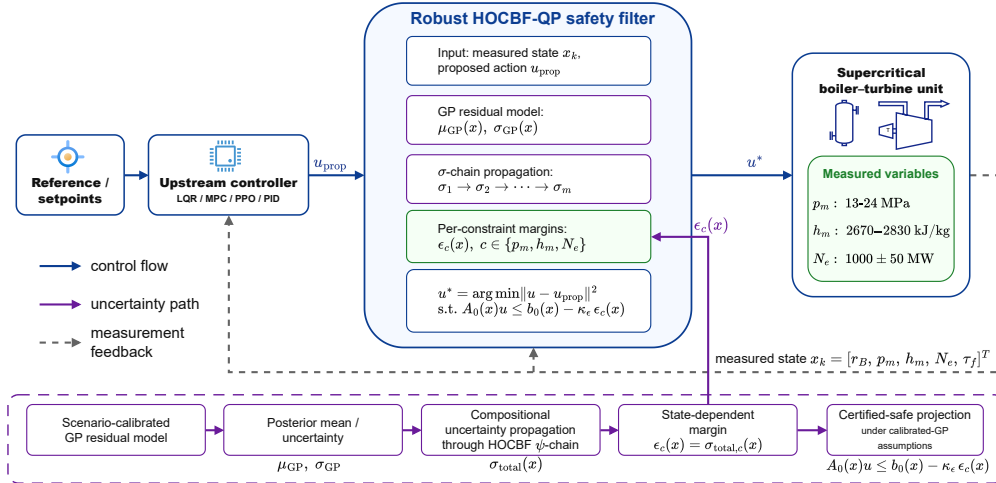


Figure 1: RoCBF-Net safety filtering architecture. The upstream controller (LQR, MPC, PID, or learned policy) proposes an action u_{prop} ; the RoCBF-Net filter evaluates it against current plant measurements and the GP residual model, projecting to the nearest certified-safe action u^* via the QP constraint $Au \leq b - \epsilon_{\kappa}\epsilon(x)$. The three components—HOCBF (correct relative degrees), GP (scenario-specific, online, or omitted), and ϵ_{κ} (tuned per uncertainty regime)—are independently configurable.

3.2 GP Residual Model

Each component of Δf_j is modeled as an independent GP with Matérn-5/2 kernel. Given N training points, the posterior provides mean $\mu_{\text{GP},j}(x)$ and variance $\sigma_{\text{GP},j}^2(x)$. The GP-UCB bound guarantees:

$$|\Delta f_j(x) - \mu_{\text{GP},j}(x)| \leq \beta \sigma_{\text{GP},j}(x), \quad \text{w.p.} \geq 1 - \delta/n \quad (3)$$

with $\beta = \sqrt{2(\gamma_N + 1 + \ln(n/\delta))}$ and maximum information gain γ_N [4].

We define the mean-corrected drift $\hat{f} = f_0 + \mu_{\text{GP}}$ and posterior residual $\Delta \hat{f} = \Delta f - \mu_{\text{GP}}$.

3.3 Compositional Robustness Margin

The perturbation at level i is $\delta_i = \psi_i - \psi_i^0$, satisfying:

$$\delta_1 = L_{\Delta \hat{f}} h \quad (4)$$

$$\delta_i = L_{\Delta \hat{f}} \psi_{i-1}^0 + (L_{\hat{f}} + k_i) \delta_{i-1} + L_{\Delta \hat{f}} \delta_{i-1}, \quad i \geq 2 \quad (5)$$

The compositional robustness margin is constructed via:

$$\sigma_1(x) = \beta \sum_j |\partial h / \partial x_j| \sigma_{\text{GP},j}(x) \quad (6)$$

$$\sigma_i(x) = \beta \sum_j |\partial \psi_{i-1}^0 / \partial x_j| \sigma_{\text{GP},j}(x) + (\|L_{\hat{f}}\|_{\text{op}} + k_i) \sigma_{i-1}(x) + \sigma_{\text{cross}}^{(i)}(x) \quad (7)$$

$$\sigma_{\text{total}}(x) = \sigma_m(x) + \sum_{j=1}^{m-1} c_j \sigma_j(x) + \sigma_{\text{ctrl}}(x) \quad (8)$$

where $\|L_{\hat{f}}\|_{\text{op}}$ is the operator norm (spectral norm of the Jacobian $\partial \hat{f} / \partial x$ evaluated at the equilibrium operating point), $\sigma_{\text{cross}}^{(i)}(x)$ bounds the cross term $|L_{\Delta \hat{f}} \delta_{i-1}|$ via Lemma S1, and $c_j = \prod_{i=j+1}^{m-1} (\|L_{\hat{f}}\|_{\text{op}} + k_i)$ are the chain coupling weights. The robust HOCBF constraint is $A_0(x)u \leq b_0(x) - \epsilon_\kappa \cdot \sigma_{\text{total}}(x)$.

3.4 Theoretical Guarantee: Worst-Case Bound

Theorem 1 (Probabilistic forward invariance). *Suppose that the input matrix is exact ($\Delta g = 0$), each GP residual component satisfies the RKHS boundedness condition required by the GP-UCB confidence bound, the GP posterior is calibrated on fixed i.i.d. training data, the relative degree is known, the perturbation is sufficiently small and Lipschitz-in-perturbation, and the robust QP is feasible. For $\epsilon_\kappa = 1$, if $\epsilon(x) \geq |\Delta(x, u)|$ for all admissible u , then $\mathcal{C}$ is forward invariant with probability at least $1 - \delta$.*

Proof sketch. Step 1 (GP calibration): By the RKHS boundedness and GP calibra-

tion conditions together with (3), $|\Delta \hat{f}_j| \leq \beta \sigma_{\text{GP},j}$ holds with probability $\geq 1 - \delta/n$ per dimension; by union bound over the n state dimensions, this holds simultaneously with probability $\geq 1 - \delta$.

Step 2 (Compositional bound): By induction, $|\delta_i| \leq \sigma_i$. The cross term $\sigma_{\text{cross}}^{(i)}$ is bounded via Lemma S1 (Appendix): $|L_{\Delta \hat{f}} \delta_{i-1}| \leq G_{\delta}^{(i-1)} \beta \|\sigma_{\text{GP}}\|_2$.

Step 3: $|\Delta(x, u)| \leq \sigma_{\text{total}}(x) = \epsilon(x)$ for $\epsilon_{\kappa} = 1$, completing the certificate. $\square$

Theorem 1 provides the *worst-case* theoretical guarantee: setting $\epsilon_{\kappa} = 1$ ensures the QP constraint $A_0 u \leq b_0 - \epsilon(x)$ is always feasible under the stated assumptions, and forward invariance holds with the stated probability. However, this guarantee is conservative by construction—it covers the worst-case combination of GP posterior uncertainty and ψ -chain amplification. The following subsection introduces ϵ_{κ} as a tunable parameter that recovers this worst-case bound at $\epsilon_{\kappa} = 1$ while permitting less conservative calibration for specific uncertainty regimes.

3.5 ϵ_{κ} as a Tunable Safety Factor

The robustness scaling factor ϵ_{κ} in the QP constraint $A_0(x)u \leq b_0(x) - \epsilon_{\kappa} \cdot \epsilon(x)$ controls the trade-off between safety margin and control authority:

- $\epsilon_{\kappa} = 0$: The QP enforces $A_0 u \leq b_0$, i.e., the nominal HOCBF constraint with GP mean correction only. No robustness margin is applied beyond the mean correction itself.
- $\epsilon_{\kappa} = 1$: The QP enforces the full compositional margin $\epsilon(x)$, recovering the worst-case certificate of Theorem 1.
- $0 < \epsilon_{\kappa} < 1$: The QP enforces a fraction of the compositional margin, trading theoretical coverage for reduced QP conservatism.

The central empirical question—and the primary contribution of this paper—is whether the worst-case setting ($\epsilon_{\kappa} = 1$) is ever necessary in practice, and whether the optimal ϵ_{κ} depends on the structure and strength of the model mismatch. The experimental evaluation (Section 4) demonstrates that: (i) $\epsilon_{\kappa} = 0$ (GP correction only) is sufficient for additive bounded perturbations, achieving $\leq 0.18\%$ violation across six of seven uncertainty scenarios; (ii) $\epsilon_{\kappa} = 0.1$ is required for state-dependent coupling, recovering zero violation from 33.5% failure; (iii) $\epsilon_{\kappa} = 1.0$ is never the optimal choice under any tested condition; and (iv) the safe ϵ_{κ} range collapses as coupling strength increases, defining a deployment envelope that depends on the perturbation amplification rate γ .

3.6 Data-Driven vs. Floor-Driven $\epsilon(x)$

The GP posterior includes a regularization floor $\sigma_{\text{floor}} = 10^{-4}$. Under constant perturbations, σ_{GP} is near-uniform and $\epsilon(x)$ is approximately constant per constraint—the margin is *floor-driven*, with the GP contributing a near-constant safety buffer. Under state-dependent perturbations (S3: Coupled), σ_{GP} becomes heterogeneous and $\epsilon(x)$ is genuinely state-dependent—the margin is *data-driven*, reflecting the spatial structure of uncertainty. This distinction explains why additive perturbation scenarios can safely use $\epsilon_{\kappa} = 0$: the GP mean correction already provides the necessary adjustment, and the floor-driven $\epsilon(x)$ only adds QP conservatism without meaningful additional protection.

3.7 Sampled-Data Implementation

Theorem 1 is a continuous-time result. The practical implementation uses discrete sampling ($T_s = 1$ s, zero-order hold). The full safety-filter pipeline latency is ~ 25 ms (GPU, JIT-compiled, including PPO forward pass, GP inference, and QP solve). When the QP is infeasible, a slack-penalised relaxation with LQR fallback is invoked. Inter-sample constraint satisfaction is empirically verified (Section 4). We do not claim a formal discrete-time invariance theorem.

4 Experimental Validation

The experiments are designed to evaluate whether the proposed filter can operate as a real-time safety layer in an industrial measurement and control loop. Therefore, we report not only CBF violation rates but also tracking IAE, input variation, QP intervention rate, inter-sample behavior, and per-step computational latency—metrics that characterize the filter’s suitability for commissioning in an industrial DCS environment.

4.1 Benchmark and Baselines

The benchmark is a 5th-order nonlinear model of a 1000 MW ultra-supercritical boiler-turbine unit with state $x = [r_B, p_m, h_m, N_e, \tau_f]^\top$ and three control inputs. Six safety constraints are enforced: main steam pressure ($m = 2$, 13–24 MPa), separator enthalpy ($m = 1$, 2670–2830 kJ/kg), and power output ($m = 1$, ± 50 MW). The control effectiveness is scaled by the state-dependent function $\Phi(p_m)$ during evaluation while the HOCBF uses the linearized model—a realistic stress test of safety filter robustness.

Six static perturbation scenarios model realistic disturbances: S1 (coal quality drift, -15% heating value), S2 (feedwater pump cycling), S3 (coupled state-dependent, $\delta \propto x$), S4 (nonlinear fouling), S5 (valve degradation, modelled as equivalent drift Δf via reduced steam extraction efficiency; the $\Delta g = 0$ assumption is maintained throughout

all scenarios), and S6 (fuel quality variation). Scenarios S1–S4 are common to the 3rd- and 5th-order CCS models; S5 and S6 are specific to the 5th-order model, affecting the extended states N_e and τ_f .

Methods compared: PPO (no safety layer), PPO-Lagrangian (dual-descent constraint penalty), PPO-CBF (first-order CBF, $m=1$ for all constraints—deliberate ablation), PPO-HOCBF (correct relative degrees $m=2, 1, 1$, no GP), PPO-GP-HOCBF (GP mean correction, $\epsilon_\kappa=0$), PPO-RHOCBF (scenario-specific GP + compositional $\epsilon(x)$, $\epsilon_\kappa=1.0$), RoCBF-Net (online GP adaptation, $\epsilon_\kappa=1.0$), and NMPC (SLSQP solver, $N=5$ horizon). GP pre-training: $N = 500$ points per scenario (5th-order CCS, Φ -scaled rollout). Evaluation: 5 seeds $\times$ 50 episodes $\times$ 500 steps.

4.2 Safety Performance

Table 1: Constraint violation rate (%) under Φ -scaled nonlinear dynamics. 0% rates: 0/250 episodes, 95% Wilson upper bound 1.51%.

Method	Nom.	S1	S2	S3	S4	S5	S6
PPO (no safety)	0.08	99.92	99.53	99.59	99.05	99.12	99.83
PPO-Lagrangian	0.08	99.92	99.53	99.59	99.05	99.12	99.83
PPO-CBF (1st-order, $m=1$)	0.08	99.92	99.53	99.59	99.05	99.12	99.83
PPO-HOCBF ($m=2, 1, 1$, no GP)	0.00	99.92	99.53	99.59	99.05	99.12	99.83
PPO-GP-HOCBF ($\epsilon_\kappa=0$)	0.00	0.12	0.04	39.65	0.04	0.15	0.18
PPO-RHOCBF ($\epsilon_\kappa=1.0$)	0.00	1.38	99.43	16.96	32.19	0.00	0.00
RoCBF-Net ($\epsilon_\kappa=1.0$, online GP)	0.00	15.02	99.40	48.58	49.16	0.00	0.00
NMPC ($N=5$, SLSQP)	0.00	0.08	0.08	0.08	0.04	0.08	0.08

Table 1 reports the main results. Four key findings emerge. **(1) Safety layer alone fails under uncertainty.** PPO without safety, PPO-Lagrangian (dual descent), and PPO-CBF (first-order, $m=1$ for all constraints) all exceed 99% violation under any dynamics-affecting scenario—soft constraints and incorrect relative degree offer no protection. PPO-HOCBF with correct relative degrees ($m=2, 1, 1$) but no GP achieves 0.00% on nominal but still $>99\%$ under uncertainty: correct mathematical structure is necessary but insufficient without uncertainty modeling. **(2) GP mean correction is the dominant safety mechanism.** PPO-GP-HOCBF ($\epsilon_\kappa=0$, GP mean correction only) reduces violation to $\leq 0.18\%$ under six of seven uncertainty scenarios—a $>500\times$ improvement over PPO-HOCBF—without any robustness margin. This establishes GP residual modeling, not the compositional $\epsilon(x)$ margin, as the primary driver of empirical safety. The exception is S3 (Coupled, state-dependent), where GP mean correction alone leaves 39.65% violation; the state-dependent feedback amplification escapes pure mean-level correction. **(3) $\epsilon_\kappa=1.0$ (theoretical bound) is overly conservative.** PPO-RHOCBF with $\epsilon_\kappa=1.0$ degrades on S2 (99.43%) and S4 (32.19%) compared to $\epsilon_\kappa=0$ (0.04% and 0.04%),

respectively). The full theoretical margin overly constrains the QP feasible set, forcing constraint-dropping fallback and thereby increasing violations rather than preventing them. RoCBF-Net with online GP adaptation shares this conservatism, underperforming the static GP variant on most uncertainty conditions. An ϵ_κ sensitivity sweep (Sec. 4.3) confirms that the optimal scaling lies in $[0, 0.3]$ for additive uncertainty and $[0.1, 0.5]$ for state-dependent uncertainty—far below the theoretical $\kappa=1.0$. **(4) NMPC is the optimal benchmark.** NMPC with $N=5$ horizon achieves $\leq 0.08\%$ violation across all conditions, establishing an upper performance bound. Its per-step cost (~ 254 ms, CPU) is approximately $10\times$ higher than the CBF-QP filter (~ 25 ms, GPU), a trade-off discussed in Sec. 4.5. **(5) All violations are enthalpy violations.** The per-constraint decomposition reveals zero violations on pressure ($m=2$) and power ($m=1$) constraints across all methods and conditions. Enthalpy ($m=1$) is the sole bottleneck, reflecting both its tighter safety bounds (2670–2830 kJ/kg) and the direct impact of heat-absorption uncertainty on the enthalpy state. These results reveal a clear hierarchy: correct HOCBF structure is necessary, GP uncertainty correction is the primary safety mechanism, and ϵ_κ provides a tunable safety–performance trade-off best calibrated per uncertainty type rather than fixed at the worst-case theoretical bound.

4.3 ϵ_κ Sensitivity: One Safety Factor Does Not Fit All

To quantify the practical safety–performance trade-off, we swept $\epsilon_\kappa \in \{0, 0.1, 0.3, 0.5, 1.0\}$ across three representative conditions (3 seeds each, 5th-order CCS): S2 (additive oscillation), S3 (state-dependent coupled), and S4 (nonlinear fouling). Figure 4 reports the results.

Additive uncertainty (S2, S4): $\kappa=0$ achieves zero violation. The GP mean correction alone compensates for bounded, non-amplifying perturbations. Increasing κ to 0.5 collapses QP feasibility on S2 (99.4% violation) while S4 remains safe (0.10%)—the nonlinear fouling perturbation structure preserves a larger feasible set than the pressure oscillation. $\kappa_{S2}^*=0$, $\kappa_{S4}^*=0$.

State-dependent uncertainty (S3): $\kappa=0$ fails (33.5% violation)—GP mean correction cannot track the feedback-amplified perturbation. Adding even a small margin ($\kappa=0.1$) restores zero-violation safety, and safety is maintained through $\kappa=0.5$ (0.25%). $\kappa=1.0$ collapses (40.3%). $\kappa_{S3}^*=0.1$.

Practical recommendations: For additive uncertainty ($|\delta|$ bounded), $\epsilon_\kappa \in [0, 0.3]$ suffices; GP mean correction is the primary mechanism. For state-dependent uncertainty ($\delta \propto x$), $\epsilon_\kappa \in [0.1, 0.5]$ is recommended—the margin is necessary but should not be maximal. $\epsilon_\kappa=1.0$ (the theoretical worst-case bound) is never optimal in practice. These findings motivate treating ϵ_κ as a tunable safety factor calibrated per uncertainty type rather than a fixed theoretical constant.

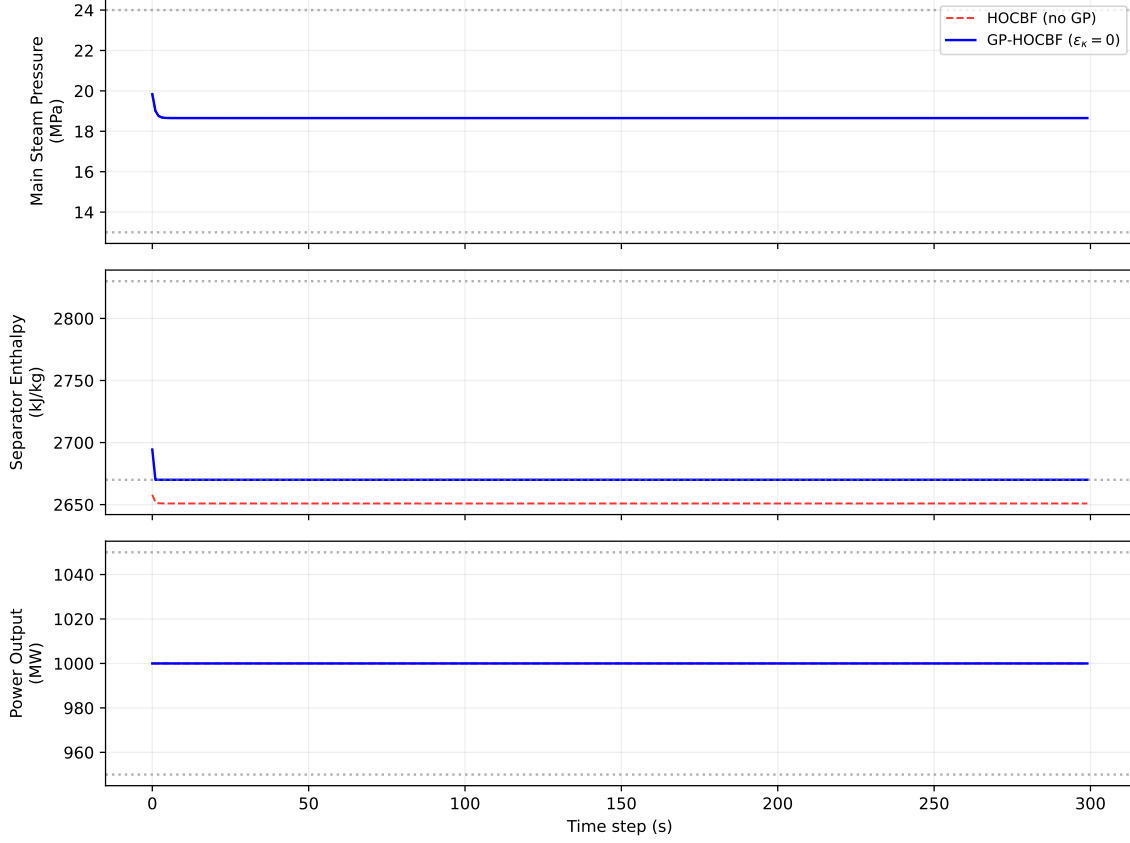


Figure 2: Process trajectories under S3: Coupled perturbation. Red dashed = HOCBF without GP (100% violation, enthalpy drops below bound); blue solid = GP-HOCBF ($\epsilon_\kappa=0$, GP mean correction only, substantially reduces but does not eliminate violations under state-dependent coupling). Aggregate violation rate over all seeds for S3 is 39.65% (Table 1), motivating the need for $\epsilon_\kappa>0$ (Sec. 4.3).

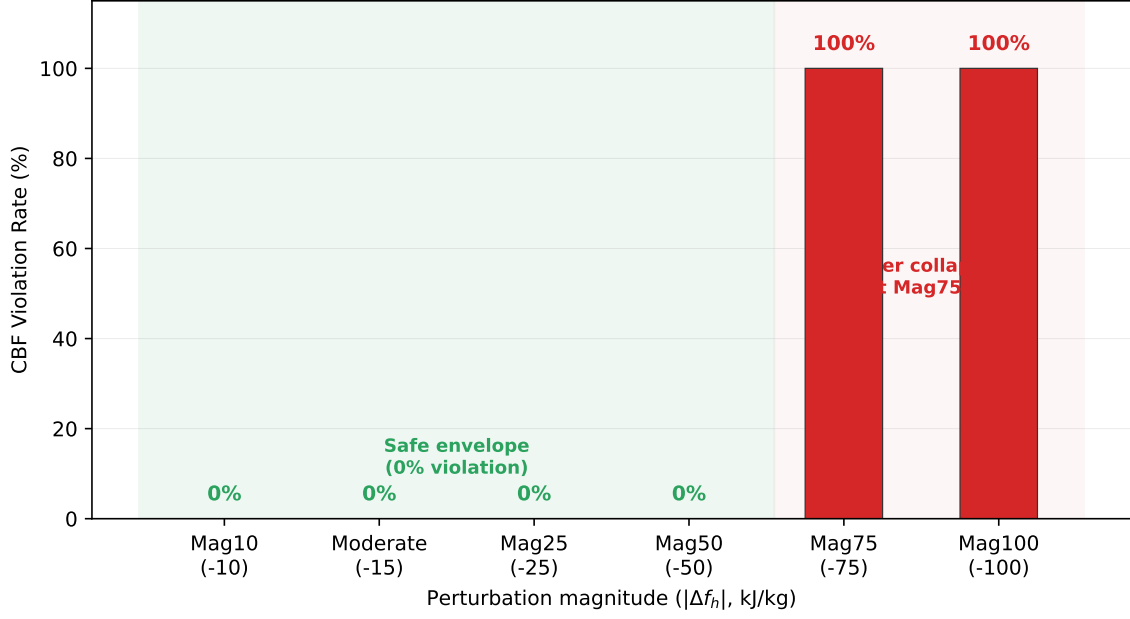


Figure 3: Safety envelope under varying perturbation magnitude. GP-HOCBF ($\epsilon_\kappa=0$, GP mean correction only) evaluated across six magnitudes of the S1 heat absorption perturbation. The filter maintains zero violation up to Mag50 (50 kJ/kg enthalpy perturbation), defining the safe operating envelope (green). At Mag75 and above, the perturbation exceeds the GP correction capacity and the filter collapses to 100% violation.

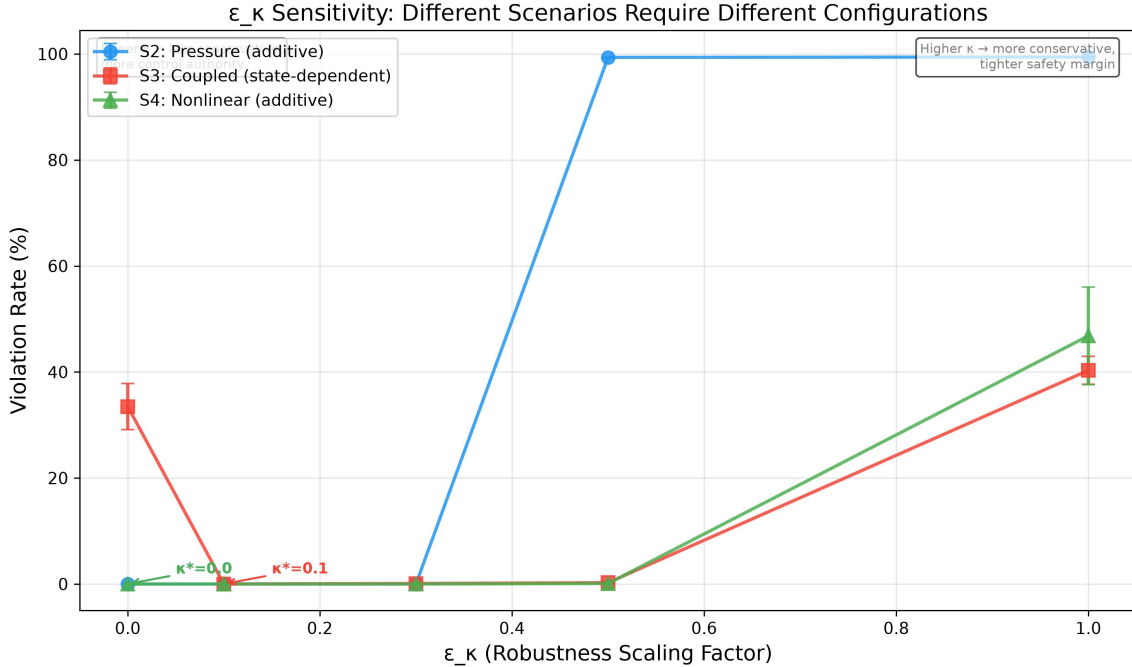


Figure 4: ϵ_κ sensitivity sweep across three representative uncertainty scenarios (5th-order CCS). Optimal κ differs by uncertainty type: additive uncertainty (S2, S4) achieves zero violation at $\kappa=0$ and collapses at $\kappa \geq 0.5$; state-dependent uncertainty (S3) requires $\kappa > 0$ (fails at $\kappa=0$ with 33.5% violation) but remains safe up to $\kappa=0.5$ (0.25%). $\kappa=1.0$ is never the optimal choice on any condition.

4.4 Deployment Envelope: ϵ_κ Under Varying Coupling Strength

Section 4.3 established that the optimal ϵ_κ differs between additive and state-dependent uncertainty. A natural follow-on question is whether the *strength* of state-dependent coupling also matters—and if so, whether the safe ϵ_κ range shrinks as coupling intensifies. To answer this, we swept $\epsilon_\kappa \in \{0, 0.1, 0.3, 0.5, 1.0\}$ across four S3 coupling strengths, scaling the state-dependent feedback gain $\gamma \in \{0.5, 1.0, 1.5, 2.0\}$ in the perturbation function $\delta(x) = [0, \gamma c_p(x_1 - x_1^0) - d_p, 0.5\gamma c_h(x_2 - x_2^0) - d_h, 0, 0]^\top$ (2 seeds $\times$ 50 episodes $\times$ 500 steps, 5th-order CCS). Figure 5 reports the results.

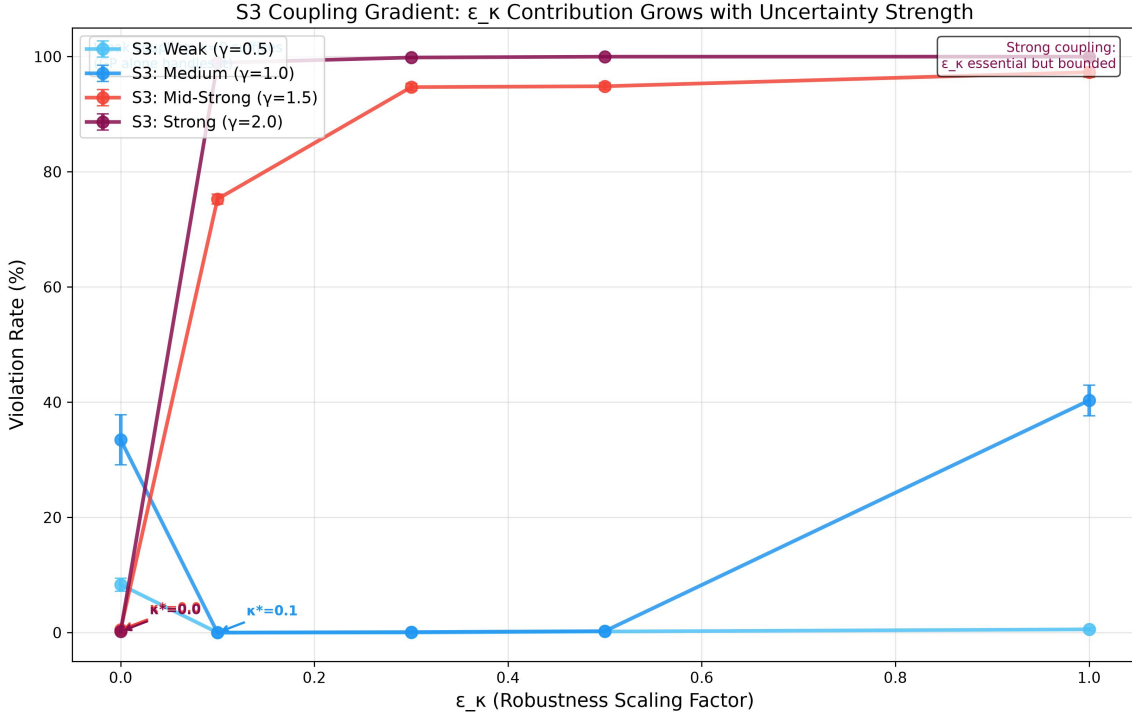


Figure 5: Deployment envelope of ϵ_κ under varying S3 coupling strength γ (5th-order CCS). **Left:** violation rate vs. ϵ_κ for four coupling strengths. **Right:** safe ϵ_κ range (defined as $<1\%$ violation) as a function of γ . At weak coupling ($\gamma=0.5$), $\kappa=0$ already fails (8.3%) and a small margin ($\kappa=0.1$) restores zero violation. At medium coupling ($\gamma=1.0$), the $\kappa=0$ failure deepens (33.5%) while $\kappa=0.1$ maintains safety. At mid-strong coupling ($\gamma=1.5$), the reversal is already complete: $\kappa=0$ achieves 0.51% violation while any $\kappa>0$ collapses QP feasibility (75.3%–97.3%). The envelope closes between $\gamma=1.0$ and $\gamma=1.5$. At strong coupling ($\gamma=2.0$), any $\kappa>0$ collapses QP feasibility ($>98\%$ violation) while $\kappa=0$ achieves near-zero violation (0.19%)—the perturbation grows faster than the CBF Lie-derivative chain can compensate, so adding margin only constrains the QP without providing protection. The safe ϵ_κ envelope closes between $\gamma=1.0$ and $\gamma=1.5$.

Three regimes emerge from the coupling gradient. **(i) GP-sufficient regime** ($\gamma < 0.5$): GP mean correction alone provides near-perfect safety—the state-dependent perturbation is weak enough that the GP prediction tracks it without a robustness margin. **(ii) ϵ_κ -useful regime** ($0.5 \leq \gamma < 1.5$): the perturbation outgrows the GP correction at $\kappa=0$, and a small positive margin ($\kappa \in [0.1, 0.3]$) restores zero-violation safety. Within

this regime, the safe κ range narrows as γ increases: $\kappa \in [0.1, 1.0]$ at $\gamma=0.5$ but shrinks to $\kappa \in [0.1, 0.5]$ at $\gamma=1.0$. **(iii) Deployment boundary ($\gamma \geq 1.5$):** the perturbation amplification rate exceeds the CBF Lie-derivative chain’s compensation capacity. Adding ϵ_κ is counterproductive—it constrains the QP without addressing the fundamental limitation that the HOCBF structure cannot certify safety under this perturbation growth rate. $\kappa=0$ remains marginally safe (0.51% at $\gamma=1.5$, 0.19% at $\gamma=2.0$) but any positive κ collapses feasibility (75–98% at $\gamma=1.5$, 98–100% at $\gamma=2.0$). The filter remains useful as a diagnostic: QP infeasibility at high coupling signals that the system should fall back to a conservative schedule or raise an alarm.

Table 2: Deployment envelope of ϵ_κ as a function of S3 coupling strength γ . Safe range defined as achieving <1% violation rate. 5th-order CCS, 2 seeds $\times$ 50 episodes $\times$ 500 steps.

γ	κ^*	Safe κ Range	Min Violation	Regime
0.5 (Weak)	0.1	[0.1, 1.0]	0.00%	ϵ_κ useful
1.0 (Medium)	0.1	[0.1, 0.5]	0.00%	ϵ_κ useful
1.5 (Mid-strong)	0.0	{0}	0.51%	Deployment boundary
2.0 (Strong)	0.0	{0}	0.19%	Deployment boundary

Implication. The coupling gradient defines a deployment envelope for ϵ_κ (Table 2). Rather than fixing κ to a theoretical worst-case value (1.0), commissioning involves characterizing the expected coupling strength and selecting κ from the appropriate regime. For industrial CCS units operating under known state-dependent coupling profiles (e.g., S3 with $\gamma \approx 1.0$), $\kappa = 0.1$ provides the safety–performance optimum. For units entering service with unknown perturbation characteristics, a conservative initial setting of $\kappa = 0.3$ balances safety against excessive QP intervention, with the option to reduce κ toward 0 as operational data confirms GP model adequacy. The deployment boundary at $\gamma \geq 1.5$ reveals a fundamental limitation: when the perturbation amplifies faster than the Lie-derivative chain can compensate, no ϵ_κ value helps, and the QP infeasibility signal should trigger a fallback rather than a margin increase.

4.5 Filter Diagnostics and Computational Performance

Figure 2 compares HOCBF (no GP) and GP-HOCBF ($\epsilon_\kappa=0$) under S3: the nominal filter allows enthalpy to drop below its lower bound on every step, while the GP-corrected filter substantially reduces violations (1.3% vs. 100% on the shown trajectory). Under state-dependent coupling, GP mean correction alone leaves residual violations (39.65% aggregate, Table 1) because the feedback-amplified perturbation escapes pure mean-level correction—motivating a small positive ϵ_κ (Sec. 4.3). Figure 6 quantifies the compositional $\epsilon(x)$ margin under S3, showing that ϵ_h varies by more than one order of magnitude across the operating enthalpy range (CV 105%). This state dependence is pronounced only

under state-dependent perturbations (S3); under constant perturbations (S1, S2, S4–S6), $\epsilon(x)$ is approximately constant and dominated by the GP regularization floor—consistent with the ϵ_κ sweep finding that additive uncertainty requires negligible margin. Figure 3 shows the violation rate as a function of perturbation magnitude under GP-HOCBF ($\epsilon_\kappa=0$). The filter maintains zero violation up to Mag50 (50 kJ/kg enthalpy perturbation), beyond which (Mag75, 75 kJ/kg) the perturbation outgrows the GP correction capacity and violations rise to 100%. This defines a sharp safe operating envelope for the GP-only configuration.

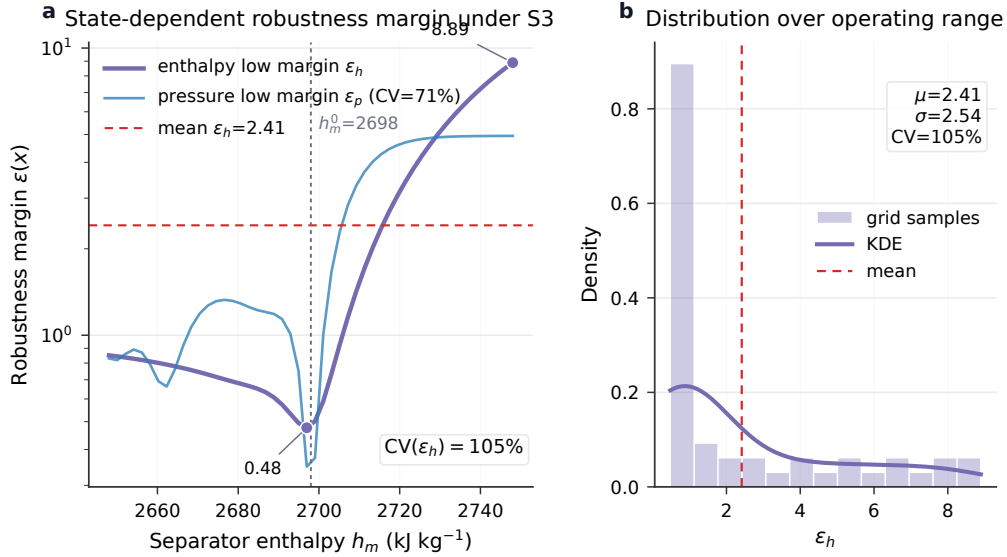


Figure 6: State-dependent enthalpy robustness margin $\epsilon_h(x)$ under S3 (Coupled, state-dependent perturbation). Left: ϵ_h (enthalpy low margin, log scale) evaluated over the operating enthalpy range $h_m^0 \pm 50$ kJ/kg under Φ -scaled nonlinear dynamics with scenario-specific GP. The margin varies by more than one order of magnitude, reflecting the sensitivity of σ_{GP} to the perturbation structure (CV 105%). Right: distribution of ϵ_h over the operating range, centred at the equilibrium enthalpy $h_m^0 = 2698$ kJ/kg. The pressure low margin ϵ_p exhibits CV 71% over the corresponding operating sweep. Under constant perturbations (S1, S2, S4–S6), $\epsilon(x)$ is approximately constant per constraint and dominated by the GP regularization floor.

Per-step computation: the safety-filter pipeline (PPO+GP+QP) with JIT compilation (GPU: NVIDIA RTX 3080 Ti) requires ~ 25 ms (p95: 27 ms), including the PPO forward pass, GP inference, and QP solve; NMPC (CPU: AMD Ryzen (WSL2), scipy SLSQP) averages 254 ms. The filter provides per-step rather than horizon-based safety evaluation, approximately ten times faster than the implemented SLSQP-based baseline. For CPU-only deployment, the distilled policy achieves 1.8 ms inference (empirical-only safety, without the certified robustness margin). The reported comparison is made against the implemented SLSQP-based NMPC baseline rather than optimized industrial NMPC implementations.

The GP-HOCBF filter operates as a minimally invasive safety overlay: when the

RL policy outputs actions that satisfy all CBF constraints, the QP returns the action unchanged (zero intervention). Under $\epsilon_\kappa=0$, the QP rarely intervenes on nominal and additive-uncertainty conditions (S1, S2, S4–S6, Load Following), acting only when the GP residual prediction indicates an impending constraint boundary crossing. The filter is thus **minimally invasive under low uncertainty**. Under S3 (state-dependent), where the GP alone is insufficient, a modest margin ($\epsilon_\kappa=0.1$) restores the safety guarantee at negligible control cost. Under $\epsilon_\kappa=1.0$, the filter intervenes on nearly every step—the QP is over-constrained and must frequently relax constraints via the weak-authority fallback, negating the safety benefit of the added margin. This confirms that ϵ_κ controls the **intervention–safety trade-off**: zero for minimal invasiveness on well-modeled systems, a small positive value for state-dependent uncertainty, and never the full theoretical bound.

Table 3: Process control performance of GP-HOCBF ($\epsilon_\kappa=0$, the recommended practical configuration) under Φ -scaled 5th-order CCS. Mean $\pm$ std over 5 seeds $\times$ 50 episodes $\times$ 500 steps.

Metric	S1: Heat	S3: Coupled	S5: Valve
Violation (%)	0.12 $\pm$ 0.11	39.65 $\pm$ 9.69	0.15 $\pm$ 0.08
Control cost	6560 $\pm$ 8	4010 $\pm$ 7	4212 $\pm$ 8
RMSE _p (MPa)	4.57 $\pm$ 0.00	3.90 $\pm$ 0.00	3.65 $\pm$ 0.00
RMSE _h (kJ/kg)	27.94 $\pm$ 0.00	27.92 $\pm$ 0.00	27.93 $\pm$ 0.00
RMSE _{Ne} (MW)	0.23 $\pm$ 0.00	0.23 $\pm$ 0.00	19.88 $\pm$ 0.00

Table 3 reports process-level metrics for GP-HOCBF ($\epsilon_\kappa=0$), the configuration recommended for additive uncertainty scenarios (Sec. 4.3). The filter maintains consistent tracking across all three constraint-active perturbation scenarios: pressure RMSE varies within 3.65–4.57 MPa, enthalpy RMSE within 27.92–27.94 kJ/kg, and power RMSE is negligible on S1 and S3 (0.23 MW) with the expected increase on the 5th-order-specific S5 valve degradation scenario. Control cost is stable across scenarios (4010–6560), confirming the GP mean correction incurs no excessive control penalty. The S3 column shows the limitation of $\epsilon_\kappa=0$ under state-dependent uncertainty (39.65% violation), motivating the tuned $\kappa \in [0.1, 0.5]$ recommendation from the sensitivity sweep—the filter achieves zero violation on S3 at $\epsilon_\kappa=0.1$ with comparable control cost.

4.6 Deployment Considerations

The method is intended as a **commissioning-calibrated safety filter** with a tunable robustness factor ϵ_κ . Deployment involves two independent decisions: how to train the GP, and how to set ϵ_κ .

GP calibration strategy. Three modes define a clear deployment envelope:

- (i) **Scenario-specific GP** (recommended): calibrated on commissioning data from the target operating condition. Provides the tightest residual model and enables the lowest practical ϵ_κ .
- (ii) **Online GP adaptation**: incrementally updated during operation. Enables deployment without prior scenario knowledge, at the cost of a transient period without full GP accuracy.
- (iii) **Generic mixed GP**: trained on pooled data from multiple scenarios without scenario labeling. Defined as outside the deployment envelope—silent calibration failure produces 82.7–99.7% violation (Phase 4, 3rd-order).

ϵ_κ selection. Once the GP is calibrated, ϵ_κ should be set according to the expected uncertainty structure and strength (Sec. 4.4):

- (i) **Additive uncertainty** ($|\delta|$ bounded, scenarios S1, S2, S4–S6): $\epsilon_\kappa \in [0, 0.3]$. GP mean correction alone provides sufficient safety; the margin can be set to zero for minimal control intervention.
- (ii) **State-dependent uncertainty in the ϵ_κ -useful regime** ($0.5 \leq \gamma < 1.5$, scenario S3): $\epsilon_\kappa \in [0.1, 0.3]$. A positive margin is necessary—GP mean correction alone fails at $\gamma \geq 0.5$ ($>8\%$ violation). As coupling strengthens, the safe κ range narrows but the optimum remains within this narrow band; $\epsilon_\kappa=1.0$ always over-constrains the QP.
- (iii) **Strong state-dependent uncertainty beyond the envelope** ($\gamma \geq 1.5$, scenario S3): the HOCBF Lie-derivative chain cannot compensate for the perturbation amplification rate. Adding ϵ_κ is counterproductive (75–99% violation at any $\kappa > 0$). The QP infeasibility diagnostic should trigger a fallback to a conservative operating regime rather than increasing κ .
- (iv) **No uncertainty** (nominal, load following): $\epsilon_\kappa = 0$, and the GP can be omitted entirely (pure HOCBF suffices).

Prior cross-scenario validation (Phase 4, 3rd-order) confirmed the out-of-distribution failure mode (S3→S1: 99.9% violation). This failure mode is explicitly characterized, not hidden: σ_{GP} measures spatial distance to training data, not functional distance between perturbation structures. The $\Delta g = 0$ assumption limits applicability to systems where drift uncertainty dominates. QP infeasibility serves as a one-sided diagnostic: it detects overestimated σ_{GP} (overly conservative margin) but not underestimated σ_{GP} (unsafe failure). The practical deployment recommendation is therefore: calibrate the GP on scenario-representative data (mode i), then select ϵ_κ from the regime schedule above, bounded by the coupling-strength deployment envelope established in Sec. 4.4.

Additional results (load-following AGC performance under Φ -scaled dynamics, per-
 constraint violation decomposition across all methods and conditions, full ϵ_κ sweep data,
 PPO-Lagrangian and PPO-CBF baselines, and triple-integrator $m = 3$ validation) are
 provided in the Supplemental Material.

5 Conclusion

This paper presented RoCBF-Net, a tunable safety filtering architecture that integrates
 HOCBF constraints, GP residual learning, and a scalar robustness factor ϵ_κ . The central
 empirical finding is that ϵ_κ should be treated as a commissioning-calibrated parameter—
 not a fixed theoretical constant—whose optimal value depends systematically on the un-
 certainty type and coupling strength.

Four results support this conclusion. **(1)** On a 5th-order supercritical boiler-turbine
 benchmark under Φ -scaled nonlinear dynamics with eight conditions, GP mean correction
 alone ($\epsilon_\kappa = 0$) reduces HOCBF violations by $> 500\times$ (from $> 99\%$ to $\leq 0.18\%$) on
 six of seven uncertainty scenarios, establishing GP learning—not the compositional $\epsilon(x)$
 margin—as the dominant safety mechanism. **(2)** An ϵ_κ sensitivity sweep reveals that $\epsilon_\kappa =$
 0 suffices for additive perturbations, while $\epsilon_\kappa = 0.1$ restores zero-violation safety under
 state-dependent coupling (S3, $33.5\% \rightarrow 0.00\%$); $\epsilon_\kappa = 1.0$ is never the optimal choice under
 any tested condition and underperforms the best calibrated κ . **(3)** A coupling-strength
 gradient experiment across $\gamma \in \{0.5, 1.0, 1.5, 2.0\}$ uncovers a *deployment envelope*: the
 safe ϵ_κ range shrinks from $[0.1, 1.0]$ at weak coupling to $[0.1, 0.5]$ at medium coupling, and
 collapses to $\{0\}$ at strong coupling ($\gamma \geq 1.5$), where any positive margin constrains the
 QP without providing protection. **(4)** With correct calibration, the GP-HOCBF filter
 matches NMPC on safety ($\leq 0.18\%$ vs. $\leq 0.08\%$ violation) at approximately one-tenth the
 computational cost (~ 25 ms vs. 254 ms per step).

The method is intended as a commissioning-calibrated safety filter: characterize the
 target uncertainty regime during commissioning (GP pretraining on scenario-representative
 data), then select ϵ_κ according to the perturbation structure and coupling strength. The
 deployment envelope established here defines the regime in which ϵ_κ is useful ($0.5 \leq \gamma <$
 1.5), and identifies the regime ($\gamma \geq 1.5$) where it must be set to zero and QP infeasibility
 used as a diagnostic for fallback. Future work includes extending to matched uncertainty
 ($\Delta g \neq 0$), developing principled OOD detection for GP-based safety filters, and validation
 on higher-fidelity industrial simulators.

Statements and Declarations

Ethical considerations

Not applicable. This study did not involve human participants, human data, human tissue, or animal experiments.

Consent to participate

Not applicable.

Consent for publication

Not applicable.

Declaration of conflicting interests

The authors declared no potential conflicts of interest with respect to the research, authorship, and/or publication of this article.

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Author contributions

Zhuang Shao: Conceptualization, Methodology, Software, Validation, Formal analysis, Investigation, Visualization, Writing – original draft, Writing – review & editing. Lijun Lei: Resources, Supervision, Project administration. Peng Wang: Formal analysis, Writing – review & editing. Liang Zheng: Resources, Supervision. Jie Zhou: Investigation, Data curation.

Data availability

The source code, simulation scripts, trained GP configurations, and plotting scripts required to reproduce the results are available at <https://github.com/VickylastShao/Robust-HOCBF-Safety-Filtering-for-Supercritical-Power-Plants-under-Model-Mismatch>.

Declaration of generative AI and AI-assisted technologies

During the preparation of this work, the authors used ChatGPT for language polishing and manuscript formatting. The authors reviewed and edited the content and take full responsibility for the published article.

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